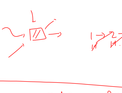
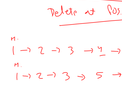


Delete at Pos

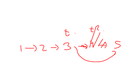
i/p: $1 \to 2 \to 3 \to 4 \to 5 \to N$, pos = 3

o/p: $1 \to 2 \to 3 \to 5 \to N$



```
t2 = t->next.
t->next = t->next->next
delete(t2)
```

```
int hops = 0;
Node* temp = head;
while (hops < pos - 1) {
    temp = temp->next;
    hops++;
}
```



pos = 3
hops = 0, 1, 2, 3

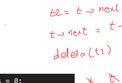
Reverse a LL

i/p: $1 \to 2 \to 3 \to 4 \to 5 \to N$

o/p: $5 \to 4 \to 3 \to 2 \to 1 \to N$

i/p: $1 \to 2 \to 3 \to 4 \to 5 \to N$

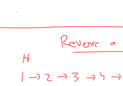
o/p: $N \leftarrow 1 \leftarrow 2 \leftarrow 3 \leftarrow 4 \leftarrow 5$



idx	node
0	1
1	2
2	3
3	4
4	5

mp[i] ->next = mp[i-1].

```
temp = curr->next.
curr->next = prev.
prev = curr.
curr = temp.
```



Head = prev

TC = O(n).
AS = O(1).

Middle

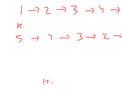
i/p: $1 \to 2 \to 3 \to 4 \to 5 \to N$

o/p: 3

i/p: $1 \to 2 \to 3 \to 4 \to 5 \to 6 \to N$

o/p: 3

Runner's Algo



Stop when f->next == NULL
for odd length.

Stop when f->next->next == NULL
for even length.

TC = O(n).
AS = O(1).

for (i: 0 → n-1)
cnt++.
for (i: 0 → n/2)
cnt++.

=

slow: 0 → n/2.
fast: 0 → n-1

for (i: 0 → n-1)
cnt++.
for (i: 0 → n-1)
cnt++.